

Arjit Praveen Kumar

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EDUCATION

BMS Institute of Technology and Management
Bachelor of Engineering in Computer Science and Engineering

Bengaluru, India
Sep. 2023 – Present

The Westminster School
IGCSE, AS & A Levels

Dubai, U.A.E
Sep. 2021 – June 2023

PROJECTS

Multithreaded HTTP Server | *C, Networking, Systems Programming* 2026

- Built an HTTP server from scratch in C using POSIX sockets, implementing the full TCP connection lifecycle including binding, listening, and accepting concurrent client connections.
- Implemented HTTP/1.1 request parsing and response formatting, serving valid responses to browsers and HTTP clients.
- Handled multiple clients concurrently using POSIX threads (pthreads).
- Managed low-level socket options such as SO_REUSEADDR and graceful connection teardown.

C2 Framework | *Python, Rust, Flask, SQLite, Adversary Emulation* 2026

- Built a command and control framework for adversary emulation with a Python/Flask server, SQLite database, and an interactive operator terminal for managing remote sessions.
- Developed a Rust beacon implant with UUID persistence, jittered sleep intervals, and HTTP task polling; cross-compiled to a Windows executable from Fedora Linux using MinGW.
- Modelled the architecture and communication protocol of real-world C2 frameworks, implementing full operator workflows for beacon registration, command tasking, and result retrieval.

CHIP-8 Emulator | *Rust, Emulation, Systems Programming* 2025

- Developed a CHIP-8 emulator in Rust with complete 35 opcodes support, and instruction decoding.
- Built core subsystems including CPU execution, memory management, timers, input handling, graphics rendering, and sound to run classic CHIP-8 programs and games while leveraging Rust's ownership model and type system to enforce memory safety.
- Tested emulator against Timendus' CHIP-8 test suite to ensure opcode and timing correctness.

CERTIFICATIONS & TRAINING

Certified Penetration Testing Specialist (CPTS) | *Hack The Box Academy* Expected 2026

SKILLS

Programming Languages: Python, Rust, Bash

Tools: Nmap, Wireshark, Burp Suite, LLMmap, PyRIT, sqlmap, John the Ripper, Hydra, Gobuster, Bloodhound, NetExec, mimikatz

Cybersecurity Concepts: Network scanning & enumeration, Web application testing, AI Red Teaming, OSINT, Active Directory, Kerberoasting, Cryptographic protocols, OWASP Top 10